AI Assisted Domestic Violence Case Pattern Recognition For Legal Aid Organizations

PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

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CaseGuard is an AI-based case management and decision-support system designed to assist legal-aid organizations, lawyers, and case workers in analyzing domestic violence cases. Such cases often involve complex statements, repeated incidents, threats, intimidation, isolation, and other forms of abuse that require careful examination before appropriate support can be provided. Manually organizing case information, identifying recurring patterns, assessing severity, and preparing legal documentation can be time-consuming and challenging. The proposed system addresses these challenges by transforming unstructured statements and case details into structured and actionable information.

The system uses Artificial Intelligence and Natural Language Processing concepts to analyze case information and identify important patterns, triggers, severity levels, and potential escalation risks. It supports the collection of structured intake forms and written statements, helping case workers organize information systematically. The analysis results provide an overview of the case and assist legal professionals in understanding recurring incidents and identifying situations that may require urgent attention.

CaseGuard provides a centralized case management dashboard where authorized users can create, view, and manage cases while monitoring their analysis status and related information. The system also generates structured lawyer briefs that summarize essential case details in an editable and printable format. These briefs reduce the manual effort involved in preparing legal documentation and help legal professionals review relevant information more efficiently.

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LIST OF ABBREVIATION

LIST OF ABBREVIATION

S.No	ABBREVIATION	FULLFORM
1	AI	ARITIFICIAL INTELLIGENCE
2	ML	MACHINE LEARNING
3	SOC	SECURITY OPERATIONS CENTER
4	NLP	NATURAL LANGUAGE PROCESSING
5	SLA	SERVICE LEVEL AGREEMENT
6	CRUD	CREATE, READ, UPDATE, DELETE
7	KYC	KNOW YOUR CUSTOMER
8	HTTP	HYPERTEXT TRANSFER PROTOCOL
9	JSON	JAVASCRIPT OBJECT NOTATION
10	RBAC	ROLE-BASED ACCESS CONTROL
11	LLM	LARGE LANGUAGE MODEL
12	CSRF	CROSS SITE REQUEST FORGERY
13	API	APPLICATION PROGRAMMING INTERFACE
14	JWT	JSON WEB TOKEN
15	DPIA	DATA PROTECTION IMPACT ASSESSMENT
16	MFA	MULTI FACTOR AUTHENTICATION

INTRODUCTION

CHAPTER 1

INTRODUCTION

CaseGuard is an AI-assisted case management and decision-support system designed to help legal-aid organizations, lawyers, and case workers organize and analyze domestic violence cases. Such cases often involve multiple incidents, threats, intimidation, coercive control, isolation, and repeated patterns of abuse. The information related to these incidents may be distributed across written statements, intake forms, timelines, and supporting documents. Manually examining and organizing this information can be time-consuming, particularly when legal professionals need to identify recurring patterns, assess the seriousness of a situation, and prepare case documentation. The proposed system aims to simplify this process by transforming unstructured case information into structured and actionable insights.

The primary motivation for CaseGuard is the difficulty of identifying meaningful patterns and potential escalation risks from complex domestic violence case records. Individual incidents may appear unrelated when examined separately, but their combined context may reveal recurring behaviour or an increasing level of concern. Reviewing these details manually requires considerable time and careful attention. CaseGuard addresses this challenge by providing a centralized platform that organizes case information and assists users in identifying relevant patterns, severity indicators, and situations that may require urgent attention.

The system uses Artificial Intelligence and Natural Language Processing concepts to examine the information provided through case intake forms and written statements. Rather than treating each document as an isolated record, the system focuses on identifying important details such as repeated incidents, threats, intimidation, isolation,

and other indicators of abusive behaviour. The extracted information is organized into a structured format to support case review and help legal professionals understand the circumstances surrounding each case. The system is intended to assist professional decision-making rather than replace legal judgment or independent risk assessment.

CaseGuard provides a centralized case management dashboard where authorized users can create, access, and manage case records. The dashboard offers a structured view of case information and analysis results, allowing legal professionals to review important details without repeatedly searching through multiple documents. By organizing case information in one place, the system aims to reduce administrative effort and improve the accessibility of relevant information during case assessment and follow-up.

The system is intended for environments where legal professionals and support organizations handle domestic violence cases and require a systematic approach to documentation and review. It can support legal-aid organizations, social support services, advocacy groups, and other authorized professionals involved in assisting affected individuals. Although the procedures followed by these organizations may differ, their common requirement is to understand the available information, identify important concerns, maintain organized records, and provide appropriate support.

An important feature of CaseGuard is its ability to connect unstructured natural language with structured case information. Statements describing incidents are generally expressed in ordinary language rather than in a predefined database format. Individuals may describe repeated threats, controlling behaviour, restrictions on communication, intimidation, or incidents occurring at different times. The system aims to organize such descriptions into meaningful information that can be reviewed systematically by authorized users.

The proposed approach considers intake forms and written statements as the primary sources of case information. The AI-assisted analysis module examines the available content and identifies relevant details, recurring patterns, and indicators associated with the reported incidents. The extracted information can then be organized to support severity assessment and the identification of possible escalation concerns. Maintaining the relationship between incidents and their surrounding context helps users review the case more comprehensively.

CaseGuard also supports the preparation of structured lawyer briefs. Legal professionals often need to review extensive case information before preparing documentation for further assessment or legal proceedings. Manually consolidating statements, incident timelines, and important observations can increase preparation time. The system aims to simplify this activity by presenting essential case details in a structured summary that can be reviewed, edited, and used as a starting point for further professional work.

The Safe Action Navigator is another important component of the proposed system. It is designed to provide relevant action guidance, evidence checklists, urgency-related information, and referrals based on the available case details and predefined rules. These features help users organize potential next steps and identify information that may be useful for seeking assistance. The guidance is intended to support informed decisions while recognizing that immediate safety needs and legal requirements must be assessed by qualified professionals.

The proposed system follows a modular architecture in which case management, information collection, AI-assisted analysis, lawyer brief generation, and action guidance operate as separate but connected components.

This modular structure makes it easier to maintain individual functions and introduce additional capabilities in the future. It also supports the systematic handling of case information throughout the process, from initial data entry to analysis, documentation, and follow-up.

Another important aspect of CaseGuard is the protection of sensitive case information. Domestic violence records may contain personal statements, incident descriptions, and other confidential details that require careful handling. The system incorporates access control and audit logging concepts to help restrict access to authorized users and maintain accountability for activities performed within the platform. These measures support responsible information management, although appropriate organizational policies and additional security safeguards remain necessary.

CaseGuard also aims to improve the efficiency of case review through structured reporting and administrative monitoring. When case details, analysis results, and supporting information are maintained systematically, authorized users can review records more easily and understand the status of their work. The organized information can support case follow-up, documentation review, and administrative oversight while reducing the need for repetitive manual data organization.

From a system perspective, the CaseGuard workflow begins with case information collection. The user enters relevant details through intake forms or provides written statements for analysis. The information is then processed by the AI-assisted analysis module, which identifies important case elements and organizes the findings into a structured representation. The resulting information is made available through the case management interface for review by authorized users.

The Case Management Module serves as the foundation for maintaining case records and associated information. The AI-Assisted Analysis Module examines the supplied content

and identifies relevant patterns and indicators. The Severity and Escalation Assessment component helps organize concerns that may require further professional attention.

The dashboard acts as the primary interface for viewing and managing case information. It enables authorized users to review records, examine analysis results, access generated briefs, and navigate available support guidance. The Administrative Monitoring and Audit Logging components provide additional oversight by helping maintain accountability and track relevant system activities.

The long-term objective of CaseGuard is to establish an intelligent and privacy-conscious case management environment that connects information collection, pattern identification, structured documentation, and action guidance within a unified workflow. By reducing the manual effort involved in organizing complex case records and supporting systematic case review, the proposed system aims to improve the efficiency of legal-aid workflows and help professionals make more informed decisions when responding to domestic violence cases.

CaseGuard further emphasizes the importance of maintaining consistency and traceability throughout the case analysis process. When information from different statements and reported incidents is organized in a structured manner, authorized professionals can compare details, review timelines, and identify information that may require clarification. Maintaining a clear connection between the original statements and the generated analysis also helps users verify the findings instead of relying entirely on automated interpretations. This approach promotes transparency and supports more reliable case documentation.

LITERATURE SURVEY

CHAPTER 2
LITERATURE SURVEY

AI-based systems for legal case management and domestic violence support have gained increasing attention due to the need for efficient case documentation, pattern identification, risk assessment, and accessible legal assistance. Domestic violence cases often involve complex narratives, repeated incidents, threats, coercive control, intimidation, and other forms of abuse that require careful examination. Traditional approaches depend heavily on manual document review and professional interpretation, which can be time-consuming when dealing with large amounts of case information. Research in Artificial Intelligence (AI), Natural Language Processing (NLP), and decision-support systems demonstrates the potential to organize unstructured narratives, extract relevant information, identify patterns, and assist professionals in reviewing sensitive cases. The CaseGuard project builds on these concepts to provide an intelligent case management platform that supports structured case analysis, severity and escalation assessment, lawyer brief generation, and safe action guidance.

2.1 AI and NLP for Domestic Violence Case Understanding

Artificial Intelligence and Natural Language Processing are increasingly used to analyze unstructured textual information and extract meaningful details from written documents. Domestic violence reports and personal statements often contain descriptions of incidents, threats, emotional abuse, controlling behaviour, and repeated patterns that may be difficult to organize manually. NLP techniques such as text classification, entity

extraction, and semantic analysis can help identify relevant information from these narratives. The CaseGuard system applies AI-assisted text analysis concepts to organize case information and identify important indicators associated with reported incidents. This helps authorized professionals review case details systematically while retaining the importance of human interpretation and contextual understanding.

2.2 Automated Incident and Pattern Extraction

Incident extraction is an important requirement in domestic violence case management because individual statements may describe multiple events occurring at different times and under different circumstances. Manual identification of these events requires careful reading and comparison of documents. Automated information extraction techniques can help identify incident descriptions, reported threats, intimidation, isolation, and recurring behaviours from textual records. The CaseGuard system aims to organize these details into structured case information so that professionals can examine reported events more efficiently. By identifying recurring elements across available statements, the system can support the review of potential patterns that might otherwise require extensive manual effort.

2.3 Severity Assessment and Escalation Risk Identification

Severity assessment and escalation risk identification are important components of domestic violence support because certain reported behaviours may indicate a need for urgent professional attention. Traditional case reviews rely on the experience and judgment of trained professionals who evaluate the circumstances, available evidence, and reported incidents. AI-assisted decision-support techniques can help organize relevant indicators and highlight information that may require further examination. The CaseGuard system supports the structured presentation of severity-related information

and potential escalation concerns based on available case details. Such analysis is intended to assist professional review rather than provide a definitive prediction of future violence or replace qualified risk assessment.

2.4 Case Documentation and Lawyer Brief Generation

Case documentation is an essential activity in legal-aid services because lawyers and case workers need organized information to understand the background, sequence, and important details of a case. Preparing summaries from multiple statements and incident records can require considerable time and may involve repetitive manual work. Automated document summarization and structured information presentation can help consolidate important details into a readable format. The CaseGuard system includes a lawyer brief generation component that organizes relevant case information into a structured summary for review and further editing. This functionality aims to reduce documentation effort while allowing legal professionals to verify the information.

2.5 Centralized Case Management and Monitoring

Centralized case management systems provide a common environment for storing, accessing, and reviewing case-related information. When documents and incident details are maintained separately, professionals may find it difficult to obtain a complete overview of a case or retrieve relevant information quickly. Digital case management approaches address this challenge by organizing records through a consistent interface. The CaseGuard system provides a centralized dashboard through which authorized users can manage case records, review analysis results, and access generated documentation. This structured environment supports improved information accessibility, systematic case review, and more efficient handling of administrative activities.

2.6 Safe Action Guidance and Referral Support

Safe action guidance is an important aspect of domestic violence support because affected individuals may require information about available assistance, documentation, and appropriate next steps. The information needed may vary depending on the reported circumstances, immediate safety concerns, and the individual's preferences. Rule-based decision-support systems can organize predefined guidance, evidence checklists, and relevant referral information in a structured manner. The CaseGuard system incorporates a Safe Action Navigator designed to present relevant action guidance and supporting checklists based on available case information and predefined rules. This component is intended to assist users in exploring appropriate support options while recognizing that individual safety decisions require careful consideration and qualified professional guidance.

2.7 Privacy, Confidentiality, and Access Control

Privacy and confidentiality are essential requirements in systems that handle domestic violence records because the information may include sensitive personal statements, incident histories, and identifying details. Unauthorized disclosure can create additional risks for affected individuals. Secure information management practices commonly include authentication, role-based access control, restricted permissions, and activity logging. The CaseGuard system incorporates access control and audit logging concepts to support responsible access to case information and maintain accountability for activities performed within the platform. These mechanisms contribute to privacy-conscious case management, while additional safeguards such as secure data storage, encryption, retention policies, and controlled information sharing are important for practical deployment.

2.8 Decision-Support Systems for Legal-Aid Organizations

Decision-support systems help professionals organize complex information and examine relevant factors before making decisions. In legal-aid environments, such systems can support document review, case prioritization, information retrieval, and the preparation of case summaries. However, legal decisions often depend on contextual details, applicable laws, professional judgment, and information that may not be available in the initial records. The CaseGuard system is designed as a decision-support platform that helps authorized professionals review organized case information and identify areas requiring further attention. By combining case management with AI-assisted analysis and structured documentation, the system aims to support legal-aid workflows without replacing professional responsibility or legal expertise.

2.9 Modular Architecture and Future Enhancements

A modular architecture is important for case management systems because different functions, including information collection, analysis, documentation, and action guidance, may require independent updates and maintenance. Separating these components allows individual modules to evolve without requiring a complete redesign of the platform. The CaseGuard system follows a modular approach that connects case intake, AI-assisted analysis, severity and escalation assessment, lawyer brief generation, the Safe Action Navigator, and administrative monitoring. This architecture provides a foundation for future enhancements, including improved document processing, incident timeline visualization, configurable assessment indicators, and integration with authorized support services. By combining modular design with privacy-conscious information management, CaseGuard aims to provide a maintainable and extensible platform for domestic violence case analysis and legal-aid support

TECH STRUCTURE

CHAPTER 3

TECHSTRUCTURE

3.1 Overview of the Technology Stack

The CaseGuard system uses a modern web technology stack designed to support domestic violence case management, AI-assisted analysis, structured documentation, and safe action guidance. The system integrates frontend technologies, backend services, database management, and analysis modules to provide a connected environment for legal-aid organizations and case workers.

- Frontend**:** Provides a user-friendly interface for creating cases, entering intake information, viewing analysis results, managing case records, and accessing lawyer briefs.
- Backend**:** Manages case-related operations, authentication, API requests, AI-assisted analysis, and communication between different system components.
- Database**:** Stores user information, case records, intake forms, statements, generated briefs, and audit logs.
- AI-Assisted Analysis**:** Processes case information to generate severity indicators, escalation scores, patterns, and triggers. The current project describes this functionality as mock ML analysis.
- Safe Action Navigator**:** Provides rule-based recommendations, evidence checklists, and referral information based on predefined rules.
- DevOps**:** Uses Docker and Docker Compose to simplify application setup and coordinate the frontend, backend, and database services.

3.2 Backend Technology

The backend of CaseGuard is developed using Node.js, Express.js, TypeScript, and Mongoose. It manages communication between the frontend, database, authentication services, and case analysis modules. The backend handles requests related to user registration, login, case creation, intake forms, statements, analysis results, lawyer briefs, and administrative operations.

The backend architecture follows a modular approach, allowing individual functionalities to be maintained and updated independently.

Key functionalities include:

- RESTful APIs**:** Handle requests related to authentication, case management, intake information, statements, generated briefs, and administrative functions.
- Case Processing**:** Receives and processes case information submitted through the frontend.
- Data Management**:** Uses Mongoose to communicate with MongoDB for storing, retrieving, and updating case records.
- AI-Assisted Analysis**:** Coordinates the analysis of case information and generates structured results, including severity indicators, escalation scores, patterns, and triggers.
- Brief Generation**:** Supports the preparation of structured lawyer briefs from available case information.
- Authentication and Authorization**:** Uses authentication middleware and role-based permissions to control access to protected operations.
- Error Handling and Validation**:** Supports request validation and consistent handling of backend errors.

3.3 Frontend Technology

The frontend of CaseGuard is developed using Next.js, React, TypeScript, Tailwind CSS, and shadcn/ui. It provides the interface through which authorized users interact with case records and access the system's different functionalities. The interface is designed to make case information easier to enter, organize, review, and manage.

Features include:

- UI Modules**:** Case management, intake forms, statement entry, analysis results, lawyer briefs, Safe Action Navigator, and administrative monitoring.
- Case Input Interface**:** Allows users to enter case details and provide information through structured forms and written statements.
- Case Management Dashboard**:** Provides a centralized view of case records and relevant information.
- Analysis Results Interface**:** Displays generated severity indicators, escalation scores, identified patterns, and triggers.
- Lawyer Brief Interface**:** Presents structured case summaries that can be reviewed, edited, and printed.
- Safe Action Interface**:** Displays rule-based guidance, evidence checklists, and referral information.
- Responsive Design**:** Uses Tailwind CSS and shadcn/ui components to support a consistent and accessible user interface across different screen sizes.

3.4 Database and Data Management

The database component of CaseGuard uses MongoDB to store and manage the information required throughout the case management process. Mongoose provides the object data modeling layer between the Node.js backend and the database.

The database maintains the information required for case creation, analysis, documentation, and administrative monitoring.

Key points:

- User Data**:** Stores user profiles, authentication-related information, and assigned roles.
- Case Data**:** Maintains case records and relevant case details.
- Intake Forms**:** Stores structured information collected during the initial case intake process.
- Statements and Narratives**:** Maintains written statements and incident-related information associated with cases.
- Analysis Data**:** Stores generated analysis results, including severity indicators, escalation scores, patterns, and triggers, where supported by the implementation.
- Lawyer Briefs**:** Maintains generated case summaries and related documentation.
- CRUD Operations**:** Supports creating, reading, updating, and deleting records according to system permissions.
- Audit Logs**:** Records relevant system activities to support accountability and administrative monitoring.

3.5 APIs and Integration

The CaseGuard system uses RESTful APIs to enable communication between the frontend, backend, database, and case analysis components. These APIs allow the user interface to submit case information and retrieve the results required for case management and documentation.

Core functionalities include:

- Case Management APIs**:** Support case creation, retrieval, modification, and other case-related operations.
- Authentication APIs**:** Handle registration, login, and authenticated access to protected resources.
- Intake and Statement APIs**:** Support the submission and management of structured forms and written narratives.
- Analysis Integration**:** Connects case information with the analysis module to produce structured case assessment results.
- Lawyer Brief Integration**:** Supports the generation, retrieval, and management of lawyer briefs.
- Safe Action Integration**:** Provides rule-based recommendations, evidence checklists, and referral information.
- Administrative Integration**:** Supports user-role management and access to relevant audit information.

3.6 Security and Authentication

Security is essential in CaseGuard because the system handles confidential domestic violence case information, personal statements, and sensitive incident records. The application includes authentication, role-based access control, and audit logging to support controlled access and accountability.

Key security features include**:**

- Authentication**:** Supports user registration and login to provide access to authorized users.
- Role-Based Access Control**:** Supports different access permissions for administrators and case workers according to system requirements.
- Password Protection**:** Uses password-hashing mechanisms to avoid storing passwords as plain text.
- Data Protection**:** Applies access restrictions to protect case information from unauthorized access.
- Input Validation**:** Uses validation mechanisms to check incoming data before processing.
- Secure API Handling**:** Uses authentication middleware and backend security measures to protect application endpoints.
- Audit Logging**:** Maintains records of relevant activities for administrative review and accountability.
- Security Middleware**:** Uses tools such as Helmet, rate limiting, and MongoDB sanitization to support application security.

3.7 Continuous Improvement and Scalability

The CaseGuard system follows a modular architecture that supports future improvements and expansion. Its separate frontend, backend, database, analysis, and guidance components allow individual features to be modified without redesigning the entire application.

- Backend Scalability**:** The backend can be extended to support additional users, case records, and organizational workflows.
- Modular Design**:** Case management, analysis, lawyer brief generation, and safe action guidance can be developed and maintained as separate components.
- Analysis Improvements**:** The existing mock ML analysis can be replaced or enhanced with validated machine-learning and NLP models as the project evolves.
- Database Expansion**:** Additional fields and data structures can be introduced to support incident timelines, evidence records, and case history.
- Deployment Support**:** Docker and Docker Compose simplify the deployment and coordination of application services.
- Future Integrations**:** The system can be extended to support authorized referral services and additional document-processing capabilities.
- Privacy and Reliability**:** Future improvements should strengthen confidentiality, access control, validation, and auditability as the system grows.

Through this technology structure, CaseGuard aims to provide a maintainable, extensible, and privacy-conscious platform for domestic violence case management and legal-aid decision support.

SYSTEM REQUIREMENT

CHAPTER 4

SYSTEM REQUIREMENT

The implementation of the CaseGuard – Domestic Violence Case Pattern Analyzer focuses on developing an intelligent, efficient, and user-friendly system for organizing domestic violence case information and supporting legal-aid workflows. The system integrates a web-based interface with backend processing, database management, AI-assisted case analysis, lawyer brief generation, and safe action guidance. The frontend provides an interactive platform where authorized users can create and manage cases, enter intake information and written statements, review analysis results, and access structured case documentation.

The backend manages system operations such as user authentication, case creation, information processing, database communication, analysis requests, and lawyer brief generation. The AI-assisted analysis module organizes case information and presents relevant patterns, severity indicators, escalation scores, and triggers. The Safe Action Navigator provides rule-based guidance, evidence checklists, and referral information. The centralized case management interface allows authorized professionals to review case information systematically while supporting confidentiality and accountability.

4.1 Hardware Requirements

The hardware requirements for the CaseGuard system include development-side and user-side computing components required for developing, testing, and operating the case management platform.

Minimum requirements include:

- Processor**:** Intel Core i5 or equivalent processor for development, application execution, and case information processing.
- RAM**:** At least 8 GB of RAM for smooth execution of the frontend, backend services, database, and development tools.
- Storage: SSD preferred for faster application execution, database operations, and access to project files.
- Display**:** Standard laptop or desktop monitor for application development and case management.
- Internet Connectivity**:** Stable internet connection for communication between the frontend, backend, database services, and other integrated components.
- Input Devices**:** Keyboard and mouse or compatible input devices for entering case information and navigating the application.
- Storage Capacity**:** Sufficient disk space for application files, database records, development dependencies, and supporting documents.

Key considerations**:**

- Processing Capability**:** Adequate processing power helps maintain smooth application performance during case creation, data retrieval, and analysis.
- Network Connectivity**:** Enables communication between authorized users and the application's services.
- Device Accessibility**:** Users can access the web application through compatible desktop computers, laptops, tablets, and smartphones.
- Data Storage**:** Sufficient storage is required for maintaining case records, intake forms, statements, generated briefs, and audit logs.

- Development Environment**:** Additional memory and processing resources may be required when running the frontend, backend, MongoDB, and Docker containers simultaneously.

The basic hardware configuration is suitable for development and prototype deployment. The system can be supported by higher-performance processors, increased RAM, and additional storage when the number of users and case records increases.

A stable network connection is important for communication between the frontend, backend, and database. Since CaseGuard handles sensitive domestic violence information, any deployment environment must also provide appropriate security measures to protect stored and transmitted data.

The system can be accessed through different user devices depending on the user's responsibilities. Administrators and legal professionals may primarily use desktop or laptop computers for detailed case review and document preparation, while other authorized users may access supported functions through mobile devices.

4.2 Software Requirements

The software environment includes the tools and technologies required for developing, testing, executing, and maintaining the CaseGuard system. The application uses web technologies, backend services, database management tools, and analysis components to coordinate case management, documentation, and safe action guidance.

The software components support communication between the user interface, backend APIs, database, authentication mechanisms, and analysis modules.

Software requirements include**:**

- Operating System**:** Windows, Linux, or macOS for application development and execution.
- Frontend Technologies**:** Next.js, React, TypeScript, Tailwind CSS, and shadcn/ui for developing the user interface.
- Backend Technologies**:** Node.js, Express.js, and TypeScript for handling application logic and API requests.
- Database: MongoDB for storing case information, user records, statements, analysis results, and audit logs.
- Database Integration**:** Mongoose for connecting the backend application with MongoDB and managing data models.
- AI-Assisted Analysis**:** The existing mock ML analysis module for generating structured analysis results. It can be extended with validated NLP or machine-learning models in future versions.
- Authentication and Security**:** Authentication middleware, role-based access control, input validation, and security middleware for protecting application resources.
- Development Tools**:** Visual Studio Code for coding, debugging, and project maintenance.
- Version Control**:** Git and GitHub for source-code management, version tracking, and collaboration.
- Deployment Tools**:** Docker and Docker Compose for configuring and running application services.
- Browser Support**:** Google Chrome, Mozilla Firefox, or Microsoft Edge for accessing and testing the web application.

4.3 Software Description

The CaseGuard – Domestic Violence Case Pattern Analyzer is a web-based case management and decision-support platform designed to organize domestic violence case information and support legal-aid workflows. The system allows authorized users to enter case details, manage intake forms and written statements, review analysis results, generate structured lawyer briefs, and access rule-based action guidance.

The frontend provides a simple and interactive interface through which authorized users can:

- Create and manage domestic violence case records.
- Enter structured case information through intake forms.
- Provide written statements and incident-related descriptions.
- View AI-assisted analysis results, including available severity indicators, escalation scores, patterns, and triggers.
- Access centralized case information through the dashboard.
- Generate, review, edit, and print structured lawyer briefs.
- View evidence checklists and rule-based guidance through the Safe Action Navigator.
- Manage authorized administrative operations and review relevant audit information.

The backend coordinates communication between the frontend, database, authentication services, and analysis components. It handles user requests, validates incoming information, manages case records, and retrieves the information required for analysis and documentation.

The database maintains user profiles, case details, intake information, statements, analysis results, generated briefs, and relevant audit records. Role-based access control helps restrict access to authorized users according to their permissions.

The AI-assisted analysis component organizes available case information and generates structured results to support professional review. The Safe Action Navigator uses predefined rules to present relevant guidance and evidence checklists. These components are intended to assist legal-aid professionals and should not replace qualified judgment, independent risk assessment, or emergency support.

The software architecture follows a modular structure in which authentication, case management, intake processing, analysis, lawyer brief generation, safe action guidance, and administrative monitoring operate as connected components. This design supports maintainability and provides a foundation for future enhancements, including improved NLP processing, incident timeline visualization, and additional authorized support-service integrations.

Overall, CaseGuard aims to reduce the manual effort involved in organizing complex case information and preparing documentation while supporting a more systematic, transparent, and privacy-conscious approach to domestic violence case management.

IMPLEMENTATION OF PROPOSED METHODOLOGY

CHAPTER 5

IMPLEMENTATION OF PROPOSED METHODOLOGY

5.1 Enhanced Case Information Processing and Pattern Analysis

CaseGuard – Domestic Violence Case Pattern Analyzer is designed to organize domestic violence case information and support the identification of important patterns through structured data processing and analysis. The system collects case details, incident descriptions, statements, timelines, and other relevant information provided by authorized users. The analysis module processes the available information to identify potential patterns, repeated incidents, threats, intimidation, isolation, and other relevant indicators. The results are presented as severity indicators and escalation-related information to help legal-aid professionals review cases more efficiently. The analysis is intended to support professional assessment and does not replace human judgment.

5.2 User-Friendly Interface

The system provides a simple and interactive user interface for managing domestic violence cases and reviewing analysis results. The dashboard includes modules for case registration, incident information, case management, pattern analysis, lawyer brief generation, and safe action guidance. Visual elements such as case status indicators, incident timelines, severity labels, and analysis summaries help authorized users understand case information easily. The interface is designed to reduce complexity and support efficient navigation for legal-aid professionals.

5.3 Automated Case Analysis and Document Generation

CaseGuard supports automated processing of submitted case information to reduce the manual effort involved in organizing and reviewing case records. The analysis module generates structured summaries of relevant incidents, potential patterns, and risk-related indicators. The system also assists in preparing lawyer briefs by organizing important case details, incident timelines, reported threats, and supporting information into a readable format. Generated briefs can be reviewed and edited by authorized professionals before being used. The current prototype uses a mock analysis module; therefore, its outputs are preliminary and should not be interpreted as validated machine-learning predictions.

5.4 Scalable and Modular Design

The system follows a modular architecture that supports maintenance, testing, and future expansion. Major modules, including user authentication, case intake, case management, pattern analysis, lawyer brief generation, the Safe Action Navigator, and administrative monitoring, are organized as separate components. These modules communicate through defined interfaces to maintain a consistent workflow. The architecture also supports future improvements, such as integrating validated natural language processing techniques, trained machine-learning models, secure document-processing services, and additional legal-aid resources.

5.5 Enhanced Security and Controlled Access

Security and confidentiality are essential because CaseGuard handles sensitive domestic violence information, personal statements, incident records, and legal case details. The system incorporates user authentication, access controls, secure data handling, and audit logging to help restrict access to authorized users and record relevant system activities. Security middleware and input validation help reduce common application risks. Access to case records and generated documents should be limited according to user roles and responsibilities. These controls are intended to protect sensitive information, although actual security depends on correct configuration, deployment, and ongoing maintenance.

5.6 Centralized Case Management and Safe Action Guidance

CaseGuard provides a centralized platform for managing domestic violence case information and supporting legal-aid workflows. Authorized users can register cases, maintain incident details, review identified patterns, generate lawyer briefs, and monitor case progress from a common dashboard. The Safe Action Navigator provides rule-based guidance, evidence checklists, urgency-related information, and relevant referral options based on the information available in the system. The guidance is intended to help users understand possible next steps while recognizing that individual circumstances vary. It does not replace advice from qualified legal professionals, trained support workers, or emergency services.

FLOWCHART

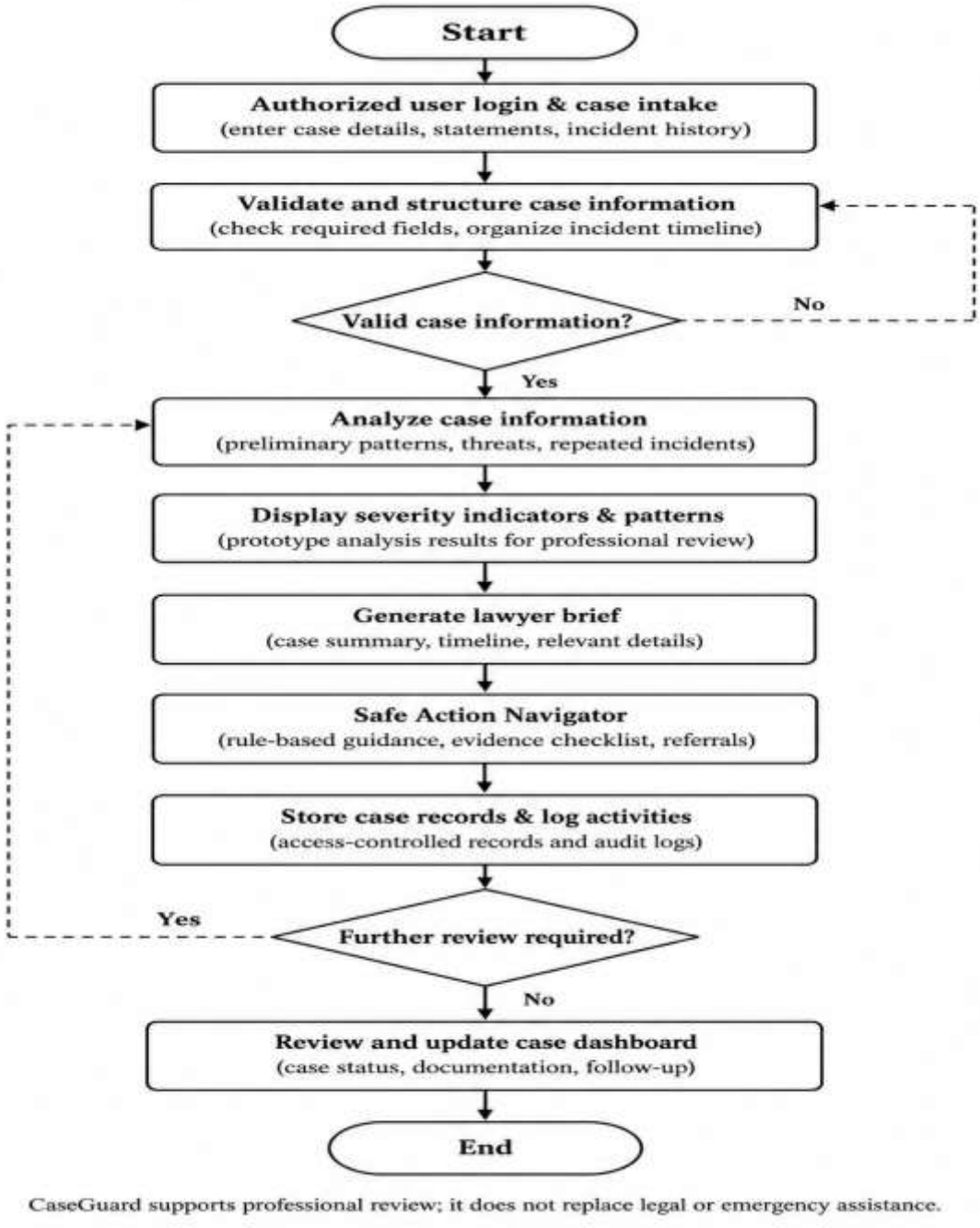


Figure 5.1 – Flowchart of the Proposed CaseGuard System

FLOW CHART DESCRIPTION

The flowchart of the CaseGuard Domestic Violence Case Pattern Analyzer represents the complete workflow of the system, from authorized user access to case analysis, documentation, and follow-up management. It illustrates how case information is collected, validated, analyzed, stored, and presented to support legal-aid professionals in reviewing domestic violence cases.

The process begins with the User Login and Case Intake Phase, where authorized users log in to the application and enter relevant case details. The information may include incident descriptions, personal statements, dates, reported threats, and previous incident history. This phase establishes the initial record required for further case processing.

After the case information is submitted, the system moves to the Case Information Validation and Structuring Phase. In this phase, the system checks the entered information for completeness and organizes the available details into a structured format. Incident dates and descriptions can be arranged chronologically to help users understand the sequence of reported events.

The system then evaluates whether the submitted case information is valid and suitable for further processing. If the information is incomplete or invalid, the workflow returns to the validation stage so that the required details can be corrected. If the information is valid, the system proceeds to the case analysis phase.

During the Case Analysis Phase, the system processes the available case information to identify potentially relevant patterns, such as repeated incidents, reported threats, intimidation, coercive control, and isolation. After processing, the system enters the Severity Indicators and Pattern Display Phase. The analysis results are presented through

the interface using severity-related indicators, potential patterns, and escalation-related information. These results are intended to support professional review. The current prototype uses a mock analysis module, so its outputs are preliminary and must not be treated as validated risk predictions.

The next stage is the Lawyer Brief Generation Phase, where important case information is organized into a structured document. The brief may contain a case overview, incident timeline, reported threats, recurring patterns, and other relevant details. This feature reduces the manual effort required to compile case information and helps legal-aid professionals prepare for further assessment.

The workflow then proceeds to the Safe Action Navigator Phase. This module provides rule-based guidance, evidence checklists, documentation suggestions, and relevant referral information based on the details available in the system. Its purpose is to help users understand possible next steps while recognizing that individual circumstances differ. The guidance does not replace professional legal advice or emergency assistance.

Finally, the system reaches the Further Review and Case Dashboard Phase, where authorized professionals determine whether additional review or analysis is required. If further review is needed, the workflow returns to the case analysis stage. Otherwise, the dashboard is updated with the latest case status, documentation, and follow-up information, completing the current case-processing cycle. Overall, CaseGuard provides a centralized platform for organizing domestic violence case information, reviewing potential patterns, preparing lawyer briefs, and supporting responsible legal-aid workflows.

RESULTS AND DISCUSSION

CHAPTER 6
RESULTS AND DISCUSSION

The CaseGuard – Domestic Violence Case Pattern Analyzer successfully demonstrates a structured approach to managing domestic violence case information and supporting legal-aid workflows. The system provides a centralized platform where authorized users can enter case details, organize incident histories, review preliminary analysis results, generate lawyer briefs, and access safe action guidance. By integrating case management, pattern analysis, documentation, and administrative monitoring, the system simplifies the organization and review of sensitive case records.

In terms of functionality, the system addresses the challenges associated with manually organizing domestic violence case information. The case intake module allows users to enter incident descriptions, statements, dates, and relevant details in a structured format. The validation and structuring process helps organize the submitted information for further analysis and review. This creates a systematic workflow from initial case registration to documentation and follow-up management.

The case analysis module supports the identification and presentation of potentially relevant patterns, including repeated incidents, reported threats, intimidation, and other behavioural indicators. The system organizes these observations into preliminary analysis results and severity-related indicators. These features can help legal-aid professionals review case information more efficiently. However, the current prototype uses a mock ML analysis module, and its outputs should not be considered validated predictions or definitive assessments of an individual's risk.

The implementation of the centralized case management dashboard improves the overall usability of the system. Authorized users can access case records, review incident details, examine analysis results, and monitor case status through a common interface. Organizing this information in one place reduces the need to search through separate documents and helps maintain a clearer overview of each case.

The lawyer brief generation functionality further improves the efficiency of case documentation. The system organizes important details, incident timelines, reported patterns, and relevant statements into a structured brief. This reduces the manual effort required to compile case information and supports preparation for legal review. The generated brief can be reviewed and edited by authorized professionals to verify its accuracy and completeness before use.

The Safe Action Navigator provides additional support by presenting rule-based guidance, evidence checklists, documentation suggestions, and relevant referral information. These features help users understand possible next steps based on the information available in the system. The guidance is intended to support informed decision-making and must be considered alongside the circumstances of each case and the advice of qualified professionals.

The system also incorporates security and administrative features to support the handling of sensitive case information. Authentication, role-based access control, and audit logging help restrict access to authorized users and maintain records of relevant system activities. These features support confidentiality and accountability when configured and maintained appropriately. Additional safeguards, including secure deployment, data retention policies, and appropriate encryption, remain important for practical use.

Overall, the prototype demonstrates how a centralized digital platform can improve the organization, accessibility, and documentation of domestic violence case information compared with a fragmented manual workflow. By combining structured case intake, preliminary pattern analysis, lawyer brief generation, rule-based guidance, and case monitoring, CaseGuard provides a systematic approach to supporting legal-aid professionals. Future enhancements could include validated NLP and machine-learning models, expert-reviewed assessment criteria, improved incident extraction, enhanced privacy controls, and usability testing with relevant professionals. These improvements would help evaluate the system's reliability and suitability for real-world legal-aid environments.

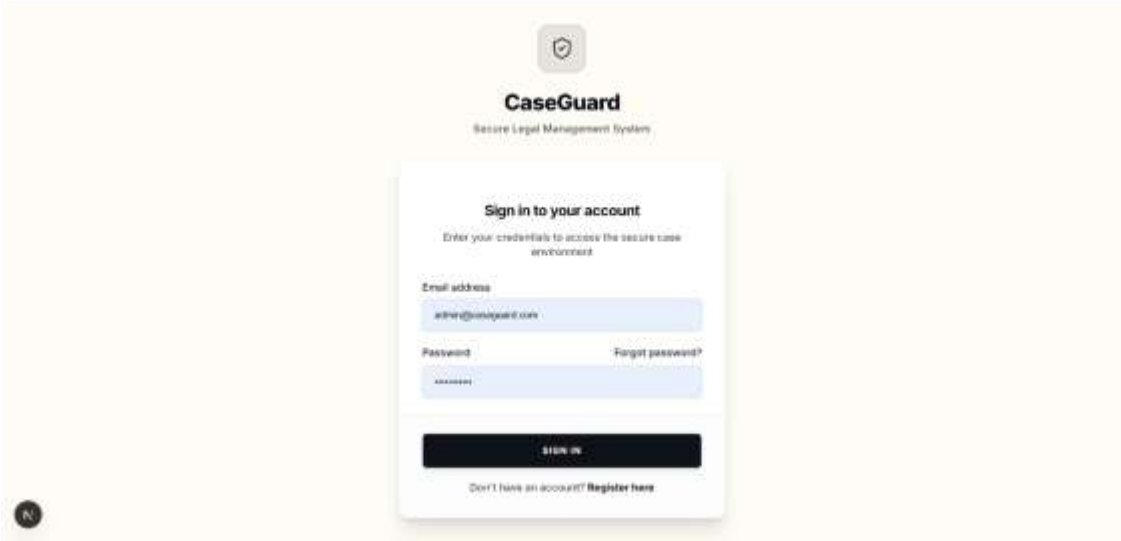


Figure 6.1Login Page

The CaseGuard login page provides secure access to the legal case management system through email and password authentication

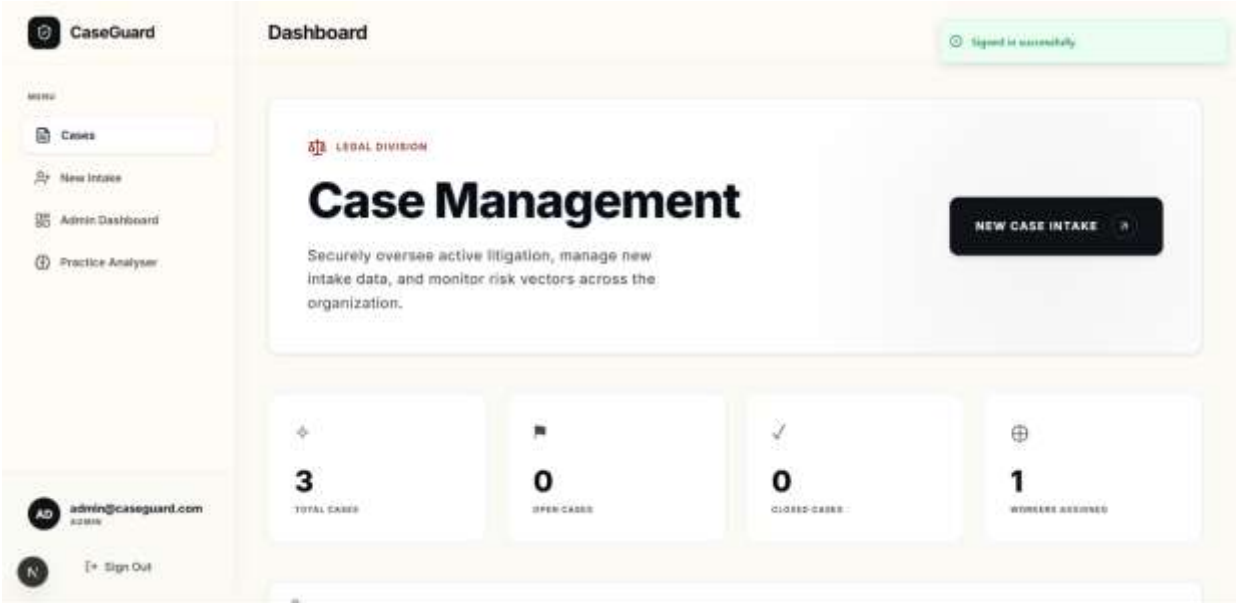


Figure 6.2 Dashboard

The CaseGuard dashboard displays case statistics and provides access to case intake, practice analysis, and administrative functions.

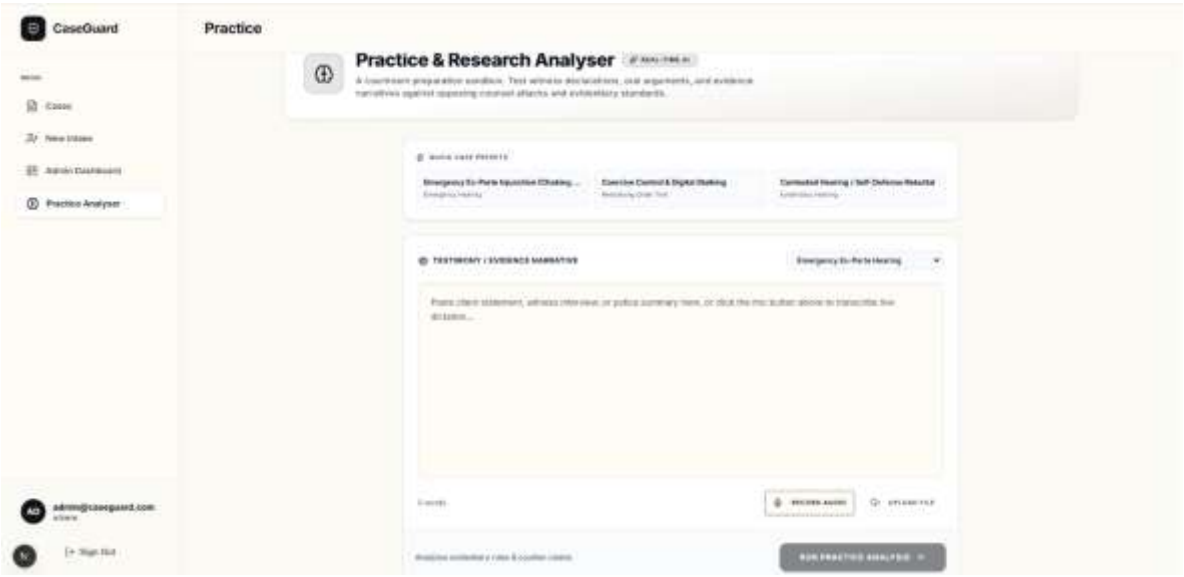


Figure 6.3 Pratice and Research Analyzer

The Practice and Research Analyzer evaluates case statements, witness testimonies, and evidence narratives to identify relevant legal considerati



Figure 6.4 Legal Document Generation

The legal document generation interface creates structured legal drafts based on case details, including statements, supporting arguments, and relevant legal information.

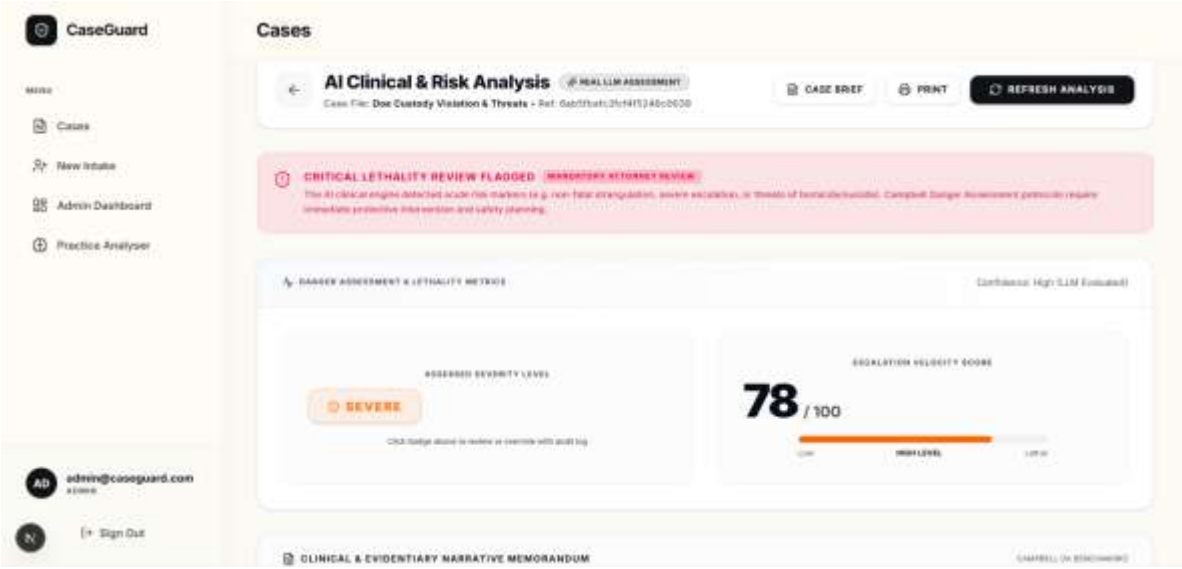


Figure 6.5 AI Clinical and Risk Analysis

The AI Clinical and Risk Analysis interface displays assessed severity level.

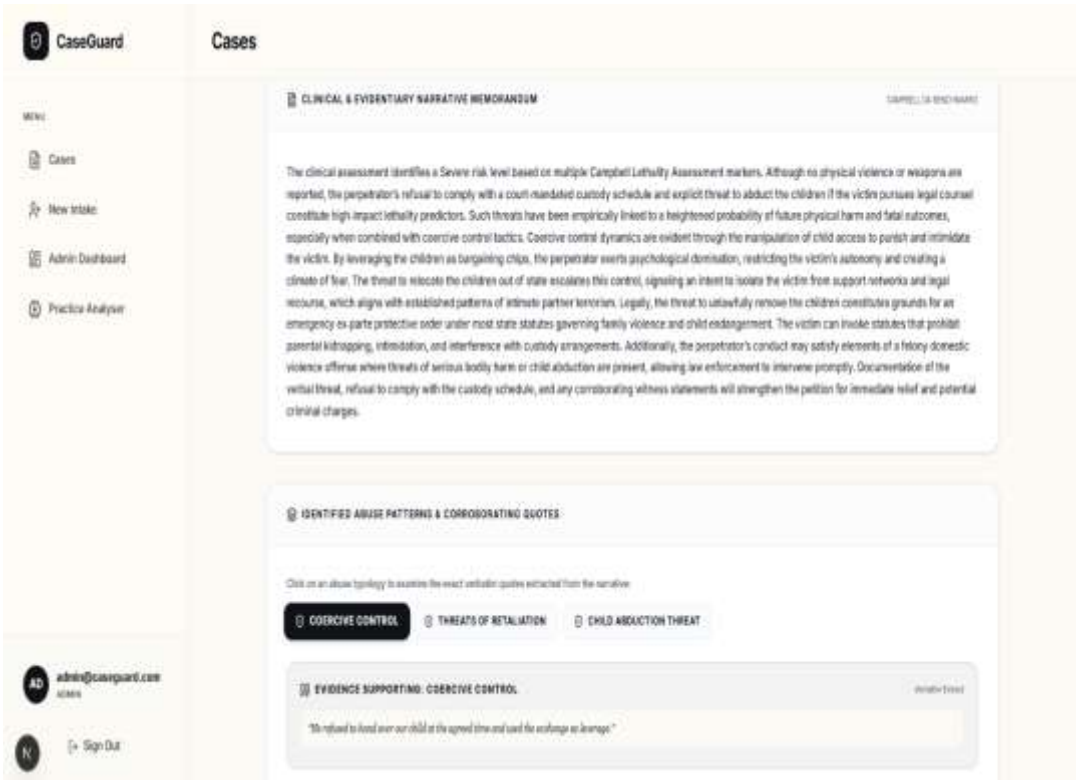


Figure 6.6 Clinical Narrative and Abuse Pattern Analysis

CaseGuard summarizes clinical risk findings and identifies abuse patterns such as coercive control, threats of retaliation, and child abduction threats with supporting evidence excerpts.

CONCLUSION AND FUTURE ENHANCEMENT

CHAPTER7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

In conclusion, the CaseGuard – Domestic Violence Case Pattern Analyzer successfully demonstrates a structured approach to managing domestic violence case information and supporting legal-aid workflows. The system aims to simplify the process of collecting, organizing, reviewing, and documenting sensitive case information through a centralized platform. By integrating case management, preliminary pattern analysis, lawyer brief generation, and safe action guidance, the system helps reduce the manual effort involved in organizing case records.

From a technological perspective, CaseGuard combines user authentication, case intake, information validation, case analysis, secure data management, lawyer brief generation, and administrative monitoring into a connected workflow. The system allows authorized users to enter incident descriptions and statements, organize incident timelines, and review preliminary indicators of potentially relevant patterns. This structured approach helps make case information easier to access and review.

The case management dashboard provides a centralized view of case records and analysis results. Authorized users can monitor case status, review incident details, and access generated documentation through a common interface. The lawyer brief generation module further supports the preparation of structured case summaries, helping legal-aid professionals organize important details for further assessment.

The Safe Action Navigator enhances the system by providing rule-based guidance, evidence checklists, documentation suggestions, and relevant referral information. Security features such as authentication, role-based access control, and audit logging support controlled access to sensitive case information. These features contribute to a more organized and accountable case management process when implemented and configured appropriately.

Overall, CaseGuard establishes a foundation for improving the organization and review of domestic violence case information. The current prototype uses a mock ML analysis module and rule-based guidance, so its analysis outputs should be treated as preliminary rather than validated risk assessments. The project demonstrates the potential of digital technologies to support legal-aid professionals while emphasizing confidentiality, professional review, and responsible handling of sensitive information.

7.2 Future Enhancement

In the future, CaseGuard can be enhanced by integrating validated Artificial Intelligence and Natural Language Processing (NLP) techniques to improve the analysis of incident descriptions and written statements. Advanced text-processing methods could help extract important details, identify recurring incidents, organize timelines, and highlight relevant information for professional review. Any such functionality should be evaluated carefully before being used in real-world case assessments.

The system can further incorporate validated methods for identifying severity-related indicators and possible escalation patterns. These methods should be developed with input from qualified domain experts and evaluated using appropriate datasets, validation procedures, and bias assessments. The objective would be to provide more reliable decision-support information without replacing professional judgment or presenting automated results as definitive conclusions.

Another important enhancement is the improvement of lawyer brief generation. Future versions could provide customizable brief templates, improved incident chronology, structured evidence references, and clearer separation between reported information and verified facts. These improvements could help legal-aid professionals prepare case documentation more efficiently while maintaining accuracy and traceability.

The Safe Action Navigator can also be expanded by incorporating verified legal-aid resources, support-service directories, and location-appropriate referral information. The system could provide more context-sensitive evidence checklists and safety-related information based on the user's needs. Such features must prioritize confidentiality and user safety, avoid assuming that a single course of action is suitable for every situation, and direct users to qualified assistance when necessary.

Security and privacy can be strengthened further through enhanced encryption, secure document handling, configurable data-retention policies, stronger access controls, and regular security testing. Additional mechanisms for managing consent, restricting access to sensitive records, and maintaining auditable access histories could improve the system's suitability for organizations handling confidential case information.

Furthermore, the analysis module can be improved through systematic testing and feedback from qualified legal-aid professionals and other relevant domain experts. Future evaluation could examine the accuracy of extracted information, the completeness of incident timelines, the usefulness of generated briefs, and the usability of the interface. Feedback should be used to identify errors and improve the system without treating user corrections alone as proof of predictive accuracy.

In the long term, CaseGuard could be developed into a more comprehensive case-support platform with improved reporting, customizable dashboards, document management, and integration with authorized legal-aid services

APPENDICES

APPENDICES

index.tsx

```
import { createFileRoute, Link } from "@tanstack/react-router";
import { Header } from "@/components/Header";
import { MeetingInput } from "@/components/MeetingInput";
import { TaskCard } from "@/components/TaskCard";
import { Dashboard } from "@/components/Dashboard";
import { Bell, CheckCircle, Clock, Users } from "lucide-react";
export const Route = createFileRoute("/")({
  head: () => ({
    meta: [
      {
        title: "MeetFlow AI — AI Meeting to Execution Operation",
      },
      {
        name: "description",
        content:
          "Transform meeting discussions into actionable tasks using AI
and Natural Language Processing.",
      },
      {
        property: "og:title",
        content: "MeetFlow AI — AI Meeting to Execution Operation",
      },
      {
```

```
      property: "og:description",
      content:
        "Convert meeting discussions into tasks, assignments,
deadlines, and execution workflows.",
    },
  ],
 }),
 component: Home,
});
function Home() {
 return (
<div className="min-h-screen bg-background">
<Header />
    {/* Navigation / Meeting Overview */}
<div className="bg-card border-b">
<div className="container mx-auto px-4 py-3 flex gap-6 overflow-
x-auto">
<Link
        to="/"
        className="text-sm font-medium text-foreground
hover:text-primary"
>
        Dashboard
</Link>
<Link
        to="/meetings"
        className="text-sm font-medium text-foreground
hover:text-primary" >Meetings
```

```
</Link>
<Link
        to="/tasks"
        className="text-sm font-medium text-foreground
hover:text-primary">
        Tasks
</Link>
<Link
        to="/notifications"
        className="text-sm font-medium text-foreground
hover:text-primary">
        Notifications
</Link>
<Link
        to="/reports"
        className="text-sm font-medium text-foreground
hover:text-primary">
        Reports
</Link>
</div>
</div> {[{
          Icon: Clock,
          title: "Pending Tasks",
          text: "Tasks awaiting execution", }, {
          Icon: CheckCircle,
          title: "Completed Tasks",
text: "Successfully completed",}, {
          Icon: Users,
```

```
            title: "Assigned Members",
            text: "Responsible team members", },{
            Icon: Bell,
            title: "Reminders",
            text: "Upcoming and overdue alerts",
          }, ].map(({ Icon, title, text }) => (
<div key={title} className="flex items-center gap-3">
<div className="h-10 w-10 rounded-full bg-primary/10 text-
primary grid place-items-center">
<Icon className="h-5 w-5" />
</div>
<div>
<p className="font-semibold text-foreground">
                {title}
</p>
<p className="text-sm text-muted-foreground">
                {text}
</p>
</div>
</div>
          ))}
</div>
</section>
<section className="container mx-auto px-4 py-8">
<MeetingInput />
</section>
<section
```

```
      id="dashboard"
      className="container mx-auto px-4 py-10">
<div className="flex items-end justify-between mb-6">
<div>
<h2 className="text-2xl font-bold text-foreground">
          Recent Action Items
</h2>
<p className="text-sm text-muted-foreground">
          AI-extracted tasks from recent meetings
</p>
</div>
</div>
<div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-
4">
<TaskCard
          title="Prepare project documentation"
          status="Pending"
          assignee="Team Member"
          deadline="Upcoming"/>
<TaskCard
          title="Review implementation progress"
          status="In Progress"
          assignee="Project Manager"
          deadline="Upcoming" />
<TaskCard
          title="Complete initial testing"
          status="Completed"
```

```
          assignee="Development Team"
          deadline="Completed" />
</div>
</section>
<footer className="bg-primary text-primary-foreground py-8 mt-10"
<div className="container mx-auto px-4 text-center text-sm opacity-90">
        © {new Date().getFullYear()} MeetFlow AI — AI Meeting to Execution Operation
</div>
</footer>
</div>);}
```

meeting.tsx

```
import { createFileRoute, useNavigate } from "@tanstack/react-router";
import { useState } from "react";
import { Header } from "@/components/Header";
import { Button } from "@/components/ui/button";
import { Textarea } from "@/components/ui/textarea";
import { Label } from "@/components/ui/label";
import { Card } from "@/components/ui/card";
import {
  FileText,
  Brain,
  CheckCircle,
  Clock,
  Users,
  ArrowRight,
```

```
} from "lucide-react";
export const Route = createFileRoute("/meetings")({
  component: MeetingPage,
  head: () => ({
    meta: [{
        title: "Process Meeting — MeetFlow AI",},  {
        name: "description",
        content:
          "Upload meeting notes or transcripts and convert discussions into actionable tasks using AI.", },],}),});
function MeetingPage()
  const navigate = useNavigate();
  const [meetingText, setMeetingText] = useState("");
  const processMeeting = () => {
    if (!meetingText.trim()) {
      return; }
    navigate({
      to: "/tasks",
    }); };
  return (
<div className="min-h-screen bg-background">
<Header />
<main className="container mx-auto px-4 py-10">
<div className="max-w-4xl mx-auto">
<div className="mb-8">
<span className="inline-block bg-primary/10 text-primary text-xs font-bold px-3 py-1 rounded-full">
```

```
            AI MEETING PROCESSING
</span>
<h1 className="text-3xl md:text-4xl font-bold mt-3">
            Convert Meeting Discussions into Actions
</h1>
<p className="text-muted-foreground mt-2">
            Provide meeting notes or transcripts and let MeetFlow AI
            identify tasks, decisions, responsible persons, and deadlines.
</p>
</div>
<Card className="p-6">
<div className="flex items-center gap-3 mb-4">
<FileText className="h-6 w-6 text-primary" />
<Label className="text-lg font-semibold">
              Meeting Notes or Transcript
</Label>
</div>
<Textarea
            value={meetingText}
            onChange={(e) => setMeetingText(e.target.value)}
            placeholder="Enter or paste the meeting notes or transcript here..."
          className="min-h-64” />
<Button
            onClick={processMeeting}
            className="mt-5"
            disabled={!meetingText.trim()}>
<Brain className="h-4 w-4 mr-2" />
```

```
            Process Meeting
    <ArrowRight className="h-4 w-4 ml-2" />
    </Button>
    </Card>
    <div className="grid md:grid-cols-4 gap-4 mt-8">
<div className="flex items-center gap-3">
    <Brain className="h-5 w-5 text-primary" />
    <span className="text-sm">AI/NLP Analysis</span>
    </div>
    <div className="flex items-center gap-3">
    <CheckCircle className="h-5 w-5 text-primary" />
    <span className="text-sm">Task Extraction</span>
    </div>
    <div className="flex items-center gap-3">
    <Users className="h-5 w-5 text-primary" />
    <span className="text-sm">Responsibility Assignment</span>
    </div>
    <div className="flex items-center gap-3">
    <Clock className="h-5 w-5 text-primary" />
    <span className="text-sm">Deadline Identification</span>
    </div>
    </div>
    </div>
    </main>
    </div>);}
```

REFERENCES

REFERENCES

[1] Campbell, J. C., Webster, D. W., and Glass, N., "The Danger Assessment: Validation of a Lethality Risk Assessment Instrument for Intimate Partner Femicide," *Journal of Interpersonal Violence*, Vol. 24, No. 4, pp. 653–674, 2009. https://doi.org/10.1177/0886260508317180

[2] Devries, K. A., et al., "The Global Prevalence of Intimate Partner Violence Against Women," *Science*, Vol. 340, No. 6140, pp. 1527–1528, 2013. https://doi.org/10.1126/science.1240937

[3] García-Moreno, C., et al., "Prevalence of Intimate Partner Violence: Findings from the WHO Multi-Country Study," *The Lancet*, Vol. 368, No. 9543, pp. 1260–1269, 2006. https://doi.org/10.1016/S0140-6736(06)69523-8

[4] Snider, C., Webster, D., O'Sullivan, C. S., and Campbell, J., "Intimate Partner Violence: Development of a Brief Risk Assessment for the Emergency Department," *Academic Emergency Medicine*, Vol. 16, No. 11, pp. 1208–1216, 2009. https://doi.org/10.1111/j.1553-2712.2009.00457.x

[5] Johnson, M. P., "Conflict and Control: Gender Symmetry and Asymmetry in Domestic Violence," *Violence Against Women*, Vol. 12, No. 11, pp. 1003–1018, 2006. https://doi.org/10.1177/1077801206293328

[6] Myhill, A., "Measuring Coercive Control: What Can We Learn from National Population Surveys?" *Violence Against Women*, Vol. 21, No. 3, pp. 355–375, 2015. https://doi.org/10.1177/1077801214568032

[7] Stith, S. M., et al., "The Intergenerational Transmission of Spouse Abuse: A Meta-Analysis," *Journal of Marriage and Family*, Vol. 62, No. 3, pp. 640–654, 2000. https://doi.org/10.1111/j.1741-3737.2000.00640.x

[8] Campbell, J. C., et al., "Risk Factors for Femicide in Abusive Relationships: Results From a Multisite Case Control Study," *American Journal of Public Health*, Vol. 93, No. 7, pp. 1089–1097, 2003. https://doi.org/10.2105/AJPH.93.7.1089

[9] Sabri, B., et al., "Assessing Risk for Severe Domestic Violence and Related Homicides Perpetrated by Partners and In-Laws," *BMC Public Health*, Vol. 24, Article 1870, 2024. https://doi.org/10.1186/s12889-024-19364-2

[10] Fawcett, T., "An Introduction to ROC Analysis," *Pattern Recognition Letters*, Vol. 27, No. 8, pp. 861–874, 2006. https://doi.org/10.1016/j.patrec.2005.10.010

[11] Breiman, L., "Random Forests," *Machine Learning*, Vol. 45, No. 1, pp. 5–32, 2001. https://doi.org/10.1023/A:1010933404324

[12] Cortes, C., and Vapnik, V., "Support-Vector Networks," *Machine Learning*, Vol. 20, No. 3, pp. 273–297, 1995. https://doi.org/10.1007/BF00994018

[13] Blei, D. M., Ng, A. Y., and Jordan, M. I., "Latent Dirichlet Allocation," *Journal of Machine Learning Research*, Vol. 3, pp. 993–1022, 2003. https://www.jmlr.org/papers/v3/blei03a.html

[14] Hochreiter, S., and Schmidhuber, J., "Long Short-Term Memory," *Neural Computation*, Vol. 9, No. 8, pp. 1735–1780, 1997. https://doi.org/10.1162/neco.1997.9.8.1735

[15] LeCun, Y., Bengio, Y., and Hinton, G., "Deep Learning," *Nature*, Vol. 521, pp. 436–444, 2015. https://doi.org/10.1038/nature14539

[16] Mikolov, T., et al., "Distributed Representations of Words and Phrases and Their Compositionality," *Advances in Neural Information Processing Systems*, Vol. 26, 2013. https://papers.nips.cc/paper/5021-distributed-representations-of-words-and-phrases-and-their-compositionality

[17] Vaswani, A., et al., "Attention Is All You Need," *Advances in Neural Information Processing Systems*, Vol. 30, 2017. https://papers.nips.cc/paper/7181-attention-is-all-you-need

[18] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *Proceedings of NAACL-HLT*, Vol. 1, pp. 4171–4186, 2019. https://doi.org/10.18653/v1/N19-1423

[19] Goldberg, Y., "A Primer on Neural Network Models for Natural Language Processing," *Journal of Artificial Intelligence Research*, Vol. 57, pp. 345–420, 2016. https://doi.org/10.1613/jair.4992

[20] Chouldechova, T., "Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments," *Big Data*, Vol. 5, No. 2, pp. 153–163, 2017. https://doi.org/10.1089/big.2016.0047